Fuel Costs, Market Power and Network Effects: A Structural Analysis of Sustainable Aviation Fuel Mandates

Shengzhuo Yuan

April 2026

Abstract

Fuel costs account for 25% to 30% of total operating costs in the airline industry, making airlines highly exposed to fuel-price shocks and climate policies that raise effective fuel costs. Recent geopolitical disruptions, including the U.S.-Iran conflict, have highlighted how fuel price spikes and volatility can lead airlines to cut flights and routes and to increase fuel surcharges and baggage fees. Despite the central role of fuel costs in the industry, little research has examined how fuel-cost shocks are transmitted into airlines' operating decisions while accounting for two key industry features: market power and network structure. This paper develops a structural model of the airline industry featuring fuel costs, market power, and network interactions. I use the model to evaluate the welfare effects of Sustainable Aviation Fuel (SAF) mandates, a leading policy tool for decarbonizing aviation. Although SAF mandates can reduce emissions, they also increase airlines' effective fuel costs and hence reduce entry, shrink network coverage, and intensify market power. The paper quantifies these tradeoffs and provides guidance for designing SAF policies that balance decarbonization, competition, and connectivity.

1 Introduction

Following the U.S.-Iran conflict that began in February 2026, crude oil prices rose sharply from \$80 per barrel to more than \$160 per barrel within about a month. The high level and substantial volatility of fuel costs have had significant effects throughout the global economy, and the airline industry has been among the sectors hit hardest. Fuel accounts for 25% to 30% of airlines' total operating costs. In response, airlines around the world have raised fares, imposed fuel surcharges and baggage fees, and reduced routes and flight frequencies. As a result, fewer carriers serve each route, potentially increasing market power. These developments have raised concerns about airline connectivity, carrier market power, and consumer welfare. Despite the importance of fuel costs for the industry, relatively little research has examined their effects while explicitly accounting for the competitive environment and the network structure of the airline industry.

The airline industry has two salient features, market power and network effects. Because of high capital costs and airport constraints, most routes are served by only a few carriers, giving rise to imperfect competition (Berry, 1992; Ciliberto and Tamer, 2009). At the same time, the hub-and-spoke network links decisions across markets. Better-connected airports can lower operating costs (Berry and Jia, 2010), while linked nonstop flights can generate one-stop itineraries without separate route-level entry (Barwick and Yuan, 2026). When market power and network effects interact, fuel-cost shocks can create localized hot spots of market power, especially at hubs where carriers control key airport infrastructure. Ignoring these industry fundamentals can therefore lead to misleading conclusions about the incidence of fuel cost shocks and the welfare effects of climate policy.

I develop a structural model of the U.S. airline industry featuring network effects, market power, and fuel costs. In the first stage, airlines choose route networks and flight frequencies to maximize expected profits. In the second stage, they compete in prices for both nonstop and one-stop products. The model captures five channels of network spillovers: 1) demand-side network benefits because passengers value airport connectivity and frequent-flyer programs; 2) marginal-cost savings in well-connected markets; 3) network effects in flight-frequency costs; 4) endogenous creation of one-stop itineraries from linked nonstop flights; 5) and network effects in route-entry fixed costs. Jet fuel prices enter the model through both the marginal cost of serving passengers and the cost of operating flights.

I then use the model to evaluate the welfare effects of sustainable aviation fuel (SAF) mandates. SAF refers to a group of low-carbon aviation fuels and is widely viewed as the most promising near-term abatement option for the airline industry. Compared with conventional jet fuel, SAF can reduce life-cycle emissions by as much as 94%, but it remains more expensive. At the same time, SAF is compatible with existing fuel infrastructure and aircraft engines, making it a realistic policy-relevant margin for the energy transition.

Several countries and regions [1] have already implemented or announced SAF policies, but their downstream welfare effects remain largely unevaluated. For example, ReFuelEU in the European Union requires the SAF blending ratio to rise from 2% in 2025 to 70% in 2050 at major EU airports. The mandate is expected to reduce aviation emissions substantially, but it also raises airlines'

[1] Besides the EU, Canada (British Columbia), India, and Brazil have announced similar SAF policy goals. The U.S., Japan, and the UK rely more on production tax credits or other forms of financial support for SAF.

effective fuel costs. In an industry characterized by thin margins, limited competition, and strong network linkages, these higher costs may reduce entry and intensify market power. The structural model developed here is designed to evaluate exactly these tradeoffs.

I plan to conduct two sets of counterfactual analyses. The first evaluates SAF mandates, modeled as increases in effective jet fuel prices. I will quantify their effects on ticket prices, flight frequencies, route networks, market power, emissions, and welfare. I will also compare alternative policy designs, including differentiated blending requirements on hub routes and alternative policy instruments such as cap-and-trade schemes, to assess whether similar emissions reductions can be achieved with smaller losses in connectivity and competition.

The second set of counterfactuals isolates the role of network effects by comparing the welfare consequences of SAF policy in models with and without network spillovers.

Preliminary results suggest that 1) higher fuel costs materially affect market structure and intensify market power; 2) airline entry decisions reflect clear network considerations; 3) the structural estimates imply economically significant network spillovers in demand, marginal costs, and flight-frequency costs.

This study contributes to three strands of literature. Firstly, it contributes to the literature on the transmission of cost shocks into firms' operating decisions, including prices and product quality (Muehlegger and Sweeney, 2022; Ruffini, 2024). In the airline industry, existing work has focused mainly on the effects of jet fuel costs on fares and service frequencies (Koopmans and Lieshout, 2016;

Sibdari, Mohammadian and Pyke, 2018). I extend this literature by studying how fuel-cost shocks affect market entry and route-network configuration in addition to fares and flight frequencies. Moreover, this study grounds the analysis on the industry fundamentals by modeling the competition environment and network effects explicitly, which are ignored in the previous literature.

Secondly, the paper adds to the literature on network effects, competition, and market power. Foundational work shows that network effects can generate positive-feedback dynamics, strengthen incumbent advantages, and make market dominance more likely by increasing the value of larger networks or installed bases (Katz and Shapiro 1994; Farrell and Saloner 1986). Recent airline network models show that ignoring network spillovers can distort welfare analysis and conceal post-merger industry dynamics (Barwick and Yuan, 2026; Bontemps, Gualdani and Remmy, 2025). I add to this literature by studying how a fuel-cost shock induced by climate policy interacts with network structure to intensify market power.

Thirdly, the paper contributes to the growing literature on SAF policy and aviation decarbonization (Wu et al., 2025; Bardon and Massol, 2025; Delbecq et al., 2023). Existing work focuses primarily on the upstream supply side and on the SAF market itself. By contrast, this study examines the downstream effects of SAF mandates in the airline product market, where policy-induced fuel-cost increases affect competition, service quality, connectivity, and welfare.

The rest of the paper is organized as follows. Section 2 provides background on the airline industry and SAF policy. Section 3 describes the data. Section 4 presents the structural model of the U.S. airline industry. Section 5 discusses identification. Section 6 reports preliminary results. Section 7 concludes and outlines the next steps of the project.

2 Airline Industry and SAF

2.1 The US Airline Industry

Since the Airline Deregulation Act of 1978, the federal government has largely withdrawn from direct control of fares, routes, and entry, allowing carriers to determine their own network and pricing decisions. Since then, the industry has evolved from largely point-to-point service toward hub-and-spoke or mixed network configurations, in part because these structures improve aircraft utilization and passenger loads.

The airline industry is therefore a natural setting in which to study the effects of fuel cost shocks. Airlines operate with thin profit margins of roughly $7 per passenger, and fuel is typically their largest operating cost, accounting for 20% to 30% of total costs. As a result, an increase in effective fuel costs can have large effects on pricing, service, and entry decisions, while the network structure can propagate those effects throughout the system.

2.2 SAF and Policy

The aviation industry accounts for roughly 3% of global carbon emissions, and this share is expected to keep rising (IATA). Aviation is a hard-to-abate sector because it relies heavily on fossil fuels and on long-lived assets and infrastructure [2]. Sustainable aviation fuel (SAF) is widely viewed as the most promising near-term decarbonization pathway for the sector. Countries and regions such as the U.S., the EU, the UK, Canada, and India have already implemented or announced policies to support SAF adoption and production. For example, ReFuelEU mandates that the SAF blending ratio rise from 2% in 2025 to 70% in 2050 at major EU airports. Despite the growing policy relevance of these mandates, there is still little empirical evidence on their overall welfare effects.

SAF refers to a class of low-carbon aviation fuels. Compared with fossil jet fuel, the life-cycle greenhouse-gas emissions of SAF are 80% to 95% lower, although its current price is still two to five times higher. Most of the emissions reduction comes from the well-to-tank stage; tailpipe emissions remain broadly similar because SAF and fossil jet fuel have similar chemical compositions. SAF is a drop-in fuel, meaning that it is compatible with existing airport infrastructure and aircraft engines and therefore does not require major retrofitting. Its production is also potentially scalable.

There are currently three main production pathways for SAF. The first is Hydroprocessed Esters and Fatty Acids (HEFA), which uses waste and residual lipids as feedstocks. HEFA is currently the least costly pathway, but it competes with other biofuels for limited feedstocks. The second is Advanced Biomass to Liquids (ABtL), which uses municipal solid waste and related feedstocks. ABtL is currently more expensive but relies on more widely available inputs. The third is Power to Liquids (PtL), which uses green hydrogen and captured carbon dioxide as feedstocks and is therefore highly scalable [3]. PtL is also currently the most expensive pathway.

[2] Aircraft can remain in service for 20 to 30 years and are costly to replace.

[3] Green hydrogen is produced through electrolysis using renewable electricity, such as solar or wind power, so feedstock availability is much less constrained.

All three pathways remain more costly than fossil jet fuel, and the cost gap is expected to persist. SAF is projected to remain roughly two to three times more expensive than fossil jet fuel through at least 2030. Johansen et al. (2026) forecast future SAF prices under alternative assumptions on hydrogen prices, technological progress, and policy support, and conclude that e-SAF produced via PtL reaches cost parity with fossil jet fuel only under the most favorable scenario and only by 2050.

3 Data

The data come from three sources. The first is the Airline Origin and Destination Survey (DB1B) from the Bureau of Transportation Statistics (BTS), which provides a 10% sample of all commercial tickets sold in each quarter. From DB1B I construct market shares, ticket fares, distances, and other product characteristics. The second source is the BTS T100 dataset, which provides flight-frequency data by aircraft type and carrier and complements the DB1B data. The third source is Annex 1.A.3.a Aviation 2023 published by the European Environment Agency (EEA), which provides estimated fuel consumption by aircraft type and distance.

Following the literature (Ciliberto and Tamer, 2009; Ciliberto, Murry and Tamer, 2021; Berry and Jia, 2010; Barwick and Yuan, 2026; Bontemps, Gualdani and Remmy, 2025), I define a market as a non-directional route between two Metropolitan Statistical Areas (MSAs) and abstract from competition among airports within the same MSA. I include the 85 largest MSAs by population, yielding $85 \times (85-1) = 7140$ potential markets. I use "MSA" and "city" inter-

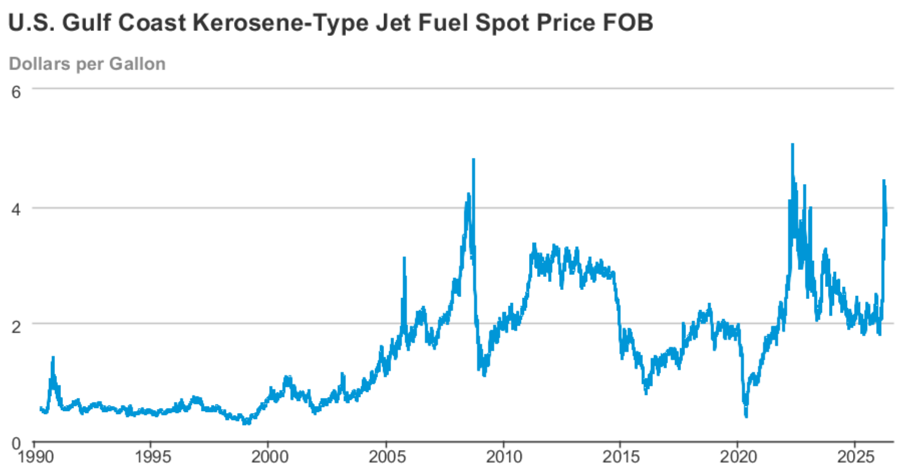

Figure 1: Jet Fuel Price

changeably throughout the paper.

The sample covers 2014:Q1 through 2018:Q4. This period contains substantial variation in jet fuel prices. Jet fuel prices fell from roughly \$2.9 to \$1.0 per gallon during 2014–2015 as in Figure 1, largely because of the shale boom, which provides a plausibly exogenous source of variation for the airline industry. Prices then fluctuated over the remainder of the sample period. In addition, American Airlines (AA) and US Airways (US) completed their merger in December 2013 after settling with the Department of Justice. Because the two carriers were already operating under common management during the sample period, I treat them as a single decision maker.

For itineraries with intermediate stops, I retain only one-stop products. Products with more than two stops account for a very small share of passengers and revenue. I define a one-stop product as in Molnar (2013): 1) the connection occurs at a hub of the operating carrier; 2) the total distance of the one-stop

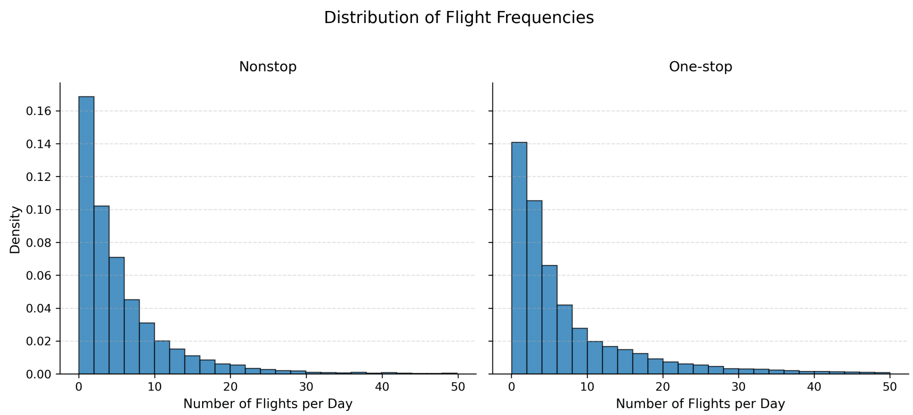

Figure 2: Density of Flight Frequencies

itinerary is less than 1.5 times the sum of the distances of the two legs; and 3) the scheduled departure time of the second flight is between 45 minutes and four hours after the arrival of the first flight. One-stop flight frequency is defined as the number of all feasible combinations of the two linked nonstop flights. The density of nonstop and one-stop flight frequencies are in Figure 2.

4 Model

The aviation industry is composed of airline companies, indexed by $n \in N$, and cities, indexed by $a, b \in C$. A market is a non-directional route between two cities and is denoted either by $m \in M$ or by ab, where $a, b \in C$. A product g is defined by its operating carrier n, its market m or ab, and, for one-stop products, its connecting city c. The model has two stages. In the first stage, airlines choose route networks and flight frequencies to maximize expected profits before observing demand and supply shocks. In the second stage, those shocks are realized and firms compete in prices for nonstop and one-stop products.

4.1 Second Stage Demand and Supply

Demand Following Berry (1994) and Bontemps, Gualdani and Remmy (2025), I adopt a nested logit model of demand. All airline products are grouped in one nest, and the outside option is placed in the other nest. The utility of passenger l from choosing product g in market m is specified as

$$U_{lgm} = b(X_{gm}, \xi_{gm}; \beta) - \alpha \times p_{gm} + \nu_{lm}(\rho) + \rho\mu_{lgm}, \tag{1}$$

where ξ_{gm} is the product-specific demand shock, ν_{lm} and μ_{lgm} are the standard nested-logit error terms, and X_{gm} includes product characteristics such as flight frequency $freq_{gm}$ and nonstop origins $ns\_origins_{gm}$, which measures how many other cities are connected to the endpoint cities of market m by nonstop flights operated by the same carrier offering product g. The term nonstop origins captures demand-side network spillovers, because passengers value airport connectedness and frequent-flyer benefits (Berry and Jia, 2010).

Then the market share is

$$s_{gm}(X_m, P_m, \xi_m; \beta, \alpha, \rho) = \frac{exp(\tau_{gm}/(1-\rho))}{exp(V_m/(1-\rho))} \times \frac{exp(V_m)}{1 + exp(V_m)}, \tag{2}$$

where $V_m = (1-\rho) \times ln(\sum_{k \in G_m} exp(\tau_{km}/(1-\rho)))$ is the expected utility of the airline product nest, $\tau_{gm} = b(X_{gm}, \xi_{gm}; \beta) - \alpha \times p_{gm}$ is the average utility of product g in market m. To compute the demand quantity, we multiply the shares with market size $q_{gm} = s_{gm} \times MS_m$.

Supply I specify the variable profit of product g in market m for firm n as

$$\pi_{ngm}(X_m, P_m, \xi_m, \omega_m; \beta, \alpha, \rho, \gamma) = \\ (p_{ngm} - mc_{ngm}(W_{ngm}, fuel\_seat_{ngm}, p_m^{fuel}, \omega_{ngm}; \gamma)) \times q_{ngm}(X_m, P_m, \xi_m; \beta, \alpha, \rho), \tag{3}$$

where $mc_{ngm}(.)$ is the marginal cost of serving passengers and depends on product characteristics W_{ngm}, the product-specific supply shock ω_{ngm}, fuel consumption per seat $fuel\_seat_{ngm}$, and the fuel price p_m^{fuel}. The term $fuel\_seat_{ngm}$ reflects aircraft fuel efficiency and captures the additional fuel cost associated with boarding extra passengers. The number of connections $conn_{ngm}$ is the sum of the nonstop origins of the endpoint cities and the intermediate stop, if any, for carrier n's product g. It captures marginal-cost-side network spillovers because densely traveled markets yield cost savings for operational reasons (Berry and Jia, 2010).

After observing the demand and supply shocks in the second stage, airlines simultaneously choose prices for all products to maximize variable profits in each market:

$$\begin{aligned} &\max_{P_n} \Pi_n(X, P, \xi, \omega; \beta, \alpha, \rho, \gamma) \\ &= \sum_{m \in M} [\max_{P_{nm}} \sum_{g \in G_{nm}} \pi_{ngm}(X_m, P_m, \xi_m, \omega_m; \beta, \alpha, \rho, \gamma) \times a_{ngm}], \end{aligned} \tag{4}$$

where M is the set of all markets, G_{nm} is the set of all potential products offered by carrier n in market m, and a_{ngm} is an indicator for whether product g of carrier n enters market m. Linked nonstop products can generate a one-stop itinerary: if $a_{nac0} = 1$ and $a_{ncb0} = 1$, then $a_{nabc} = 1$, where endpoint city pairs are used to label markets. The terms $a_{nac0} = 1$ and $a_{ncb0} = 1$ denote nonstop products in markets ac and cb provided by carrier n, while $a_{nabc} = 1$ denotes a one-stop product in market ab with an intermediate stop at city c.

4.2 First Stage Entry and Flight Frequencies

Flight Frequency Cost Each carrier can offer at most one nonstop product in a market, so flight-frequency costs can be indexed by the carrier-market pair nm or nab. I use endpoint city pairs to label markets in order to distinguish more clearly between nonstop and one-stop products. For example, f_{nab0} denotes the number of nonstop flights operated by carrier n in market ab, while f_{nabc} denotes the number of one-stop flights in market ab connecting through city c and operated by carrier n. I assume that the flight-frequency cost for nonstop service in market ab is linear in the number of flights f_{nab0}

$$\Gamma_{nab}(f_{nab0}, W^f_{nab0}, fuel_{nab0}, p^{fuel}_{ab}, \epsilon_{nab}; \delta) = c^f(W^f_{nab0}, fuel_{nab0}, p^{fuel}_{ab}, \epsilon_{nab}; \delta) \times f_{nab0}, \tag{5}$$

where $c^f(.)$ is the marginal cost of operating a flight and depends on product characteristics W^f_{nab0}, fuel consumption per flight $fuel_{nab0}$, and the fuel price p^{fuel}_{ab}. Nonstop origins also enter W^f_{nab0} and hence provide another channel of network effects.

I specify the fixed cost as follows

$$FC_n(A_n, \kappa_n; \eta) = \sum_{ab \in M} a_{nab} \times (\eta_{1n} + \kappa_{nab}) + \sum_{h \in H_n} \eta_{2n} (\sum_{a \in C, a \neq h} a_{nah})^2, \tag{6}$$

where H_n is the set of carrier n's hubs. The first term is the linear entry cost. The second, quadratic term captures congestion costs at hubs and therefore constitutes another channel of network effects.

Airlines choose route networks and flight frequencies to maximize expected profits:

$$\begin{aligned} \max_{A_n^@, F_n^@} & E[\Pi_n(A^@, X^@, P^@, \xi^@, \omega^@; \beta, \alpha, \rho, \gamma) | A^@, X^@, P^@] \\ & - \Gamma_n(A_n, F_n, W_n^f, Fuel_n, P_{fuel}, \epsilon_n; \delta) \\ & - FC_n(A_n, \kappa_n; \eta), \end{aligned} \tag{7}$$

where $V^@$ denotes the set of variables V for all potential products, including those not chosen by the carriers.

4.3 Technical Relationship of One-stop Flight Frequency

Computing one-stop flight frequencies directly in the counterfactual analysis would be computationally infeasible because it would require modeling the exact departure and arrival time of each flight. I therefore adopt an alternative mapping from nonstop flight frequencies to one-stop flight frequencies.

Following Barwick and Yuan (2026), I specify the number of one-stop flights in market ab connecting through city c as a function of the number of nonstop flights in markets ac and cb:

$$f_{nabc} = \psi_{nabc}(f_{nac0}, f_{nbc0}), \tag{8}$$

where the technical relationship ψ_{nab} is carrier and market specific. A change in nonstop flight frequency in market ac can affect one-stop flight frequencies in as many as $2 \times (C-2)$ markets, where C is the number of cities. This reflects the network spillovers from flight frequencies.

I will explore alternative functional forms for this relationship. Barwick and Yuan (2026) find that a symmetric Cobb-Douglas specification fits the data well.

4.4 Equilibrium

The Subgame Perfect Nash Equilibrium is $\{A^*, F^*, P_m^*(A_m^@, F_m^@, \xi_m^@, \omega_m^@)\}$, including the first stage optimal route network and flight frequency choices, and the second stage contingent pricing equation. The pricing equation is contingent on market product decisions, flight frequencies, and demand, supply shocks $P_m^*(A_m^@, F_m^@, \xi_m^@, \omega_m^@)$, which is the solution of the optimal variable profits in the 2nd stage as in Equation 4. The optimal route network (A^*) and flight frequency (F^*) choices solve Equation 7.

Nocke and Schutz (2018) establish existence and uniqueness of the second-stage pricing equilibrium for the multi-product nested logit model. Bontemps, Gualdani and Remmy (2025) show that existence of the first-stage equilibrium is much harder to establish. Following Barwick and Yuan (2026) and Bontemps, Gualdani and Remmy (2025), I assume that a first-stage equilibrium exists. The first-stage estimation relies on inequality methods as in Pakes (2010) and Pakes et al. (2015) and therefore does not require equilibrium uniqueness.

5 Identification

This section presents the identification strategy for the second-stage demand parameters (β, α, and ρ), the second-stage supply parameters (γ), and the first-stage flight-frequency and entry-cost parameters (δ and η).

5.1 Identification of Second Stage Parameters

I follow the identification strategy in Berry (1994) and Berry, Levinsohn and Pakes (1995) to estimate the second-stage parameters. There are two main sources of endogeneity. First, the observed products are selected by carriers and may therefore be correlated with the second-stage shocks. Second, prices and within-nest shares are endogenous in the nested logit model.

I assume that the second-stage shocks are not included in the carrier's first-stage information set, that is, ξ and ω are mean independent of the first-stage information set. Formally, $E[\xi_{gm}, \omega_{gm} | X^{@}, W^{@}, \epsilon, \kappa] = 0$ for $\forall g \in G_m$. This assumption is standard in the literature. In this setting it is a reasonable simplification because the model already controls for several forms of network interaction. The mean-independence assumption addresses the first source of endogeneity and allows me to instrument for prices and within-nest shares in the usual way following Berry (1994).

Identificaiton of Demand Following Berry (1994), the nested-logit market share can be written in closed form as

$$lns_{ngm} - lns_{0m} = b(X_{ngm}, \xi_{ngm}; \beta) - \alpha \times p_{ngm} + \rho \times lns_{ngm|o}, \qquad (9)$$

where s_{0m} is the share of the outside good and $s_{ngm|o}$ is the within nest share of product g. I parameterize $b(X_{ngm}, \xi_{ngm}; \beta) = X_{ngm}\beta + \xi_{ngm}$. Both the prices and within nest shares are endogenous. I instrument for them using standard BLP-style instruments, including sums of competing-product characteristics such as flight frequencies and nonstop origins, as well as the number of nonstop and one-stop products within the nest.

Identification of Supply Following Berry, Levinsohn and Pakes (1995), I recover marginal costs from the F.O.C. of the optimal pricing problem in Equation 4. The marginal cost can then be estimated as

$$\hat{mc}_{nm} = p_{nm} + [\frac{\partial s_{nm}(X_m, P_m, \xi_m; \beta, \alpha, \rho)}{\partial p_{nm}}]^{-1} s_{nm}(X_m, P_m, \xi_m; \beta, \alpha, \rho), \quad (10)$$

where all terms on the right-hand side of Equation 10 are observed or estimated. This inversion yields estimates of the marginal cost of serving passengers. I parameterize the marginal cost function as $mc(W_{ngm}, fuel\_seat_{ngm}, p_m^{fuel}, \omega_{ngm}; \gamma) = W_{ngm}\gamma_1 + \gamma_2 \times p_m^{fuel} \times fuel\_seat_{ngm} + \omega_{ngm}$. I then regress the estimated marginal cost $\hat{mc}$ on the parameterized cost shifters to recover γ.

5.2 Identification of First Stage Parameters

With the second stage estimates, I can simulate the variable profits for different network and flight frequency choices and carry out the first stage estimation.

Identification of Marginal Cost of Flight Frequency Recall the optimization problem in the first stage from Equation 7, the optimal flight frequencies are given by the F.O.C. of Equation 7 with respect to flight frequencies. The F.O.C. with respect to the nonstop flight frequency in market ab is

$$
\begin{aligned}
&\frac{\partial E[\Pi_n(A^{@}, X^{@}, P^{@}, \xi^{@}, \omega^{@}; \beta, \alpha, \rho, \gamma)|A^{@}, X^{@}, P^{@}]}{\partial f_{nab0}} \\
&= \frac{\partial \Gamma_n(A_n, F_n, W_n^f, Fuel_n, P_{fuel}, \epsilon_n; \delta)}{\partial f_{nab0}} \\
&= c^f(W_{nab0}^f, fuel_{nab0}, p_{ab}^{fuel}, \epsilon_{nab}; \delta),
\end{aligned} \tag{11}
$$

where the left hand side can be computed with data and estimates from the second stage. I provide an anatomy of it as follows

$$
\begin{aligned}
&\frac{\partial E[\Pi_n(A^{@}, X^{@}, P^{@}, \xi^{@}, \omega^{@}; \beta, \alpha, \rho, \gamma)|A^{@}, X^{@}, P^{@}]}{\partial f_{nab0}} \\
&= \frac{\partial E(\pi_{nab0})}{\partial f_{nab0}} \\
&+ \sum_{c \in C_{nab}, c \neq 0} \frac{\partial E(\pi_{nabc})}{\partial f_{nab0}} \\
&+ \sum_{k \in C_{nb}, k \neq a} \frac{\partial E(\pi_{nakb})}{\partial f_{nab0}} + \sum_{k \in C_{na}, k \neq b} \frac{\partial E(\pi_{nbka})}{\partial f_{nab0}} \\
&+ \sum_{k \in C_{nb}, k \neq a} \sum_{c \in C_{nak}, c \neq b} \frac{\partial E(\pi_{nakc})}{\partial f_{nab0}} + \sum_{k \in C_{na}, k \neq b} \sum_{c \in C_{nbk}, c \neq a} \frac{\partial E(\pi_{nbkc})}{\partial f_{nab0}},
\end{aligned} \tag{12}
$$

where C_{na} refers to all cities connected with a by nonstop flights operated by carrier n, and C_{nab} refers to the set of intermediate stops for all one-stop products of carrier n in market ab. This decomposition highlights four components of the marginal benefit of higher flight frequency: 1) the profit gain from the nonstop product in market ab itself; 2) the cannibalization of existing one-stop products in market ab; 3) the profit gain from one-stop products that use market ab as one of their links; and 4) the cannibalization of existing products in markets whose one-stop products also use market ab as a link. All of these terms can be computed using the data and the second-stage estimates [4].

[4] I compute the expectation by taking 200 random draws of the second stage demand and

I parameterize the frequency cost $c^f(W^f_{nab0}, fuel_{nab0}, p^{fuel}_{ab}, \epsilon_{nab}; \delta) = W^f_{nab0}\delta_1 + delta_2 \times p^{fuel}_{ab} \times fuel_{nab0} + \epsilon_{nab0}$. I then estimate δ by OLS using the recovered flight-frequency costs and the corresponding shifters in Equation 12.

Identification of Entry Fixed Cost In the presence of network effects, computing equilibrium network structures directly is extremely difficult, if not impossible, because entry or exit in one market can affect profits in as many as $2 \times (C-2)+1$ other markets. I therefore estimate entry costs using the revealed-preference approach of Pakes (2010) and Pakes et al. (2015). The key intuition is that observed entry decisions provide lower and upper bounds on entry costs. For example, if a carrier enters market m, then the increase in expected variable profits from entry must exceed the increase in entry fixed costs, which provides an upper bound on entry cost. Conversely, the decision not to enter provides a lower bound.

To formalize this identification strategy, I introduce additional notation. Let A^*_n denote the observed network structure. Let $A^a_n(-ab)$ denote an alternative network structure in which the entry status of some markets for carrier n is changed. By revealed preference,

$$\Delta\Pi_n(A^a_n, A^*_n; \eta | A^*_{-n}) = \Pi_n(A^a_n; \eta | A^*_{-n}) - \Pi_n(A^*_n; \eta | A^*_{-n}) \leq 0 \qquad (13)$$

Let us break this down into two terms. One contains expected variable profits and flight-frequency costs, and the other contains entry fixed costs.

supply shocks, ξ and ω, and then take the average of implied flight marginal cost.

$$\Delta\Pi_n(A_n^a, A_n^*; \eta | A_{-n}^*) = \Delta\Pi_n^1(A_n^a, A_n^*; \eta | A_{-n}^*) + \Delta\Pi_n^2(A_n^a, A_n^*; \eta | A_{-n}^*) + \phi_n^a(\kappa_n) \tag{14}$$

$$\begin{aligned}\Delta\Pi_n^1&(A_n^a, A_n^*;\eta | A_{-n}^*)\\ &= \{E[\Pi_n(A_n^a, F_n^a) - \Pi_n(A_n^*, F_n^*)]\} - [\Gamma_n(A_n^a, F_n^a) - \Gamma_n(A_n^a, F_n^a)]\end{aligned} \tag{15}$$

$$\begin{aligned}\Delta\Pi_n^2&(A_n^a, A_n^*;\eta | A_{-n}^*)\\ &= -\{\sum_{ab \in M} \eta_{1n} \times (a_{nab}^a - a_{nab}^a) + \sum_{h \in H_n} \eta_{2n}[(\sum_{a \in C, a \neq h} a_{nah}^a)^2 - (\sum_{a \in C, a \neq h} a_{nah}^*)^2]\}\end{aligned} \tag{16}$$

The term $\Delta\Pi_n^1(A_n^a, A_n^*; \eta | A_{-n}^*)$ in Equation 15 depends only on data and previously estimated parameters and is therefore computable. By contrast, $\Delta\Pi_n^2(A_n^a, A_n^*; \eta | A_{-n}^*)$ in Equation 16 depends on the unknown entry-cost parameters.

The error term $\phi_n^a(\kappa_n)$ in Equation 14 is endogenous and does not have mean zero because carriers observe the entry cost shocks κ_n when making network decisions. I therefore use instruments J that are uncorrelated with $\phi_n^a(\kappa_n)$. Candidate instruments include population-quantile dummies and gate-ownership-quantile dummies (Barwick and Yuan, 2026), as well as historical entry-status profiles (Bontemps, Gualdani and Remmy, 2025). The resulting moment inequality is

$$E[J \times \Delta\Pi_n(A_n^a, A_n^*; \eta | A_{-n}^*)] + E[J \times \phi_n^a(\kappa_n)]$$
$$= E[J \times \Delta\Pi_n(A_n^a, A_n^*; \eta | A_{-n}^*)] \leq 0. \quad (17)$$

I compute the sample analog of the moment inequalities in Equation 17 and estimate the model following Pakes et al. (2015). For computational feasibility, I plan to focus on alternative network structures that differ from the observed network by only one market due to computational feasibility.

6 Preliminary Results

In this section, I present preliminary results, including reduced-form evidence on entry patterns and structural estimates for the second stage and for flight-frequency costs in the first stage.

6.1 Reduced Form Evidence

As shown in Figure 3, I plot passenger and revenue shares across market structures for the bottom and top quartiles of jet fuel prices. More concentrated market structures – monopoly, duopoly, and triopoly markets – account for larger shares of passengers and revenue when jet fuel prices are high than when they are low. This pattern suggests that higher fuel costs intensify market power.

I next provide more formal evidence that jet fuel prices affect market structure and market power. In Table 1, I estimate a linear probability model with an entry indicator as the dependent variable. Because of the high-dimensional

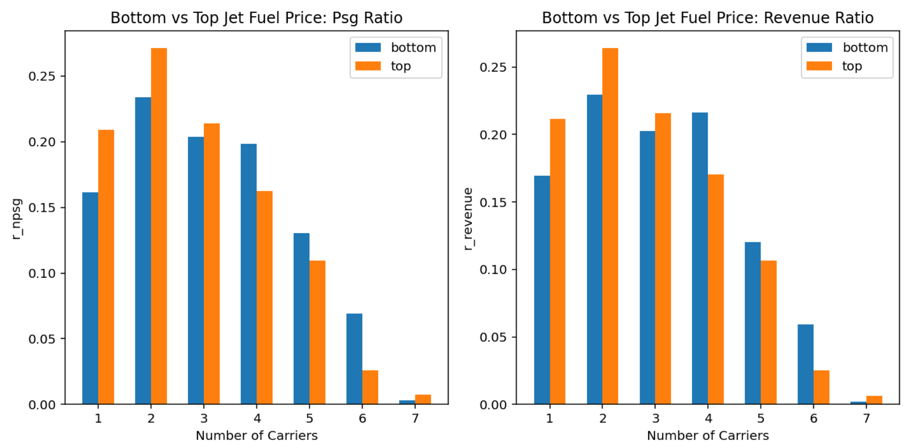

Figure 3: Passenger and Revenue Ratio by Market Structure

fixed effects, I do not estimate logit or probit models. In addition to fuel costs, I include measures of route centrality for both the carrier and its competitors. I measure centrality using two metrics, nonstop origins as in Berry and Jia (2010) and PageRank as in Lawrence et al. (1999). The two measures deliver similar results. I use lagged centrality because entry decisions are simultaneous within a period.

Four findings emerge: 1) a \$1 increase in jet fuel price per gallon reduces entry probabilities by 9% to 13%; 2) the fuel-cost effect on entry is stronger on more central routes; 3) centrality encourages entry; and 4) airlines avoid competitors' most central routes.

6.2 Estimates for Technical Relationship of One-stop Flights

Table 2 reports estimates for the relationship between one-stop flight frequency and the sum of nonstop frequencies on the two linked markets. The R^2 is high, indicating that the specification fits the data well. The results are robust across alternative fixed-effect specifications.

Table 1: Reduced Form Entry Model

	Entry	Entry
distance	0.00081^{***}	0.001^{**}
	(0.00031)	(0.00047)
distance2	$3.4e-07^{***}$	$7.3e-07^{***}$
	(6.8e-08)	(9.4e-08)
Fuel Price	-0.087^{**}	-0.13^{***}
	(0.036)	(0.033)
L.Ori NS Origins	0.0094^{***}	
	(0.00017)	
L.Dest NS Origins	0.0085^{***}	
	(0.00019)	
L.NS Origins Cmpt	-0.0064^{***}	
	(5.5e-05)	
Fuel Price * L.Ori NS Origins	-0.00054^{***}	
	(8.1e-05)	
Fuel Price * L.Dest NS Origins	-0.00038^{***}	
	(9.4e-05)	
L.Ori PagrRank		0.0049^{***}
		(0.00011)
L.Dest PageRank		0.003^{***}
		(0.00011)
L.Ori PageRank Cmpt		-0.01^{***}
		(0.00018)
L.Dest PageRank Cmpt		-0.011^{***}
		(0.00025)
Fuel Price * L.Ori PageRank		-0.00044^{***}
		(5.2e-05)
Fuel Price * L.Dest PageRank		-0.00031^{***}
		(5.4e-05)
Firm FEs	YES	YES
Route FEs	YES	YES
Quarter FEs	YES	YES

We regress entry dummy on different independent variables, including origin and destination cities' centrality measure, jet fuel price, and their interaction terms. We also include centrality measure of competitors. We measure centrality using two metrics: 1) Nonstop Origins as in Berry and Jia (2010); 2) PageRank as in Lawrence et al. (1999). The results show that high fuel prices discourage entry, high centrality measures encourage entry and airlines avoid "central" routes of competitors. These are consistent with the network spillovers and the industry fact of a thin profit margin and high fuel cost shares. The firm FEs, route FEs and quarter FEs are absorbed. Standard errors are clustered at market level. Due to the high dimension of FEs, we don't adopt a Logit/Probit model, instead we adopt a Linear Probability Model.

Table 2: Technical Relationship between Nonstop and Onestop Flight Frequency

	log(nflight)	log(nflight)	log(nflight)
lnflights12	0.99***	0.98***	0.99***
	(0.0027)	(0.0028)	(0.0035)
_cons	−7.15***	−6.97***	−7.17***
	(0.036)	(0.037)	(0.047)
Firm FEs	NO	YES	YES
Route FEs	NO	NO	YES
Quarter FEs	YES	YES	YES
R-square	0.76	0.77	0.83
N	59204	59204	59155

We estimate the relationship between the one-stop flights and sum of nonstop flights of its two links.

6.3 Structural Estimates

I present the structural estimates of the model in section 4. The results indicate economically meaningful network spillovers.

Second Stage Demand and Supply Estimates As shown in Table 3, passengers value nonstop service, higher flight frequencies, and better-connected networks. These findings support the presence of demand-side network spillovers.

Table 4 reports second-stage supply estimates. More densely connected markets, measured by the number of connections, exhibit lower marginal costs, consistent with marginal-cost-side network spillovers. Nonstop products also have lower marginal costs than one-stop products. Higher flight frequencies reduce marginal costs as well, which is intuitive because more frequent service provides more options for passengers and reduces the need to add another flight to accommodate incremental demand.

The estimated coefficient on $p_m^{fuel} \times fuel\_seat_{ngm}$ is about 0.25. This term

Table 3: Demand Estimates in the Second Stage

	NL	NL	NL
distance	0.0044***	0.0061***	0.0041***
	(0.00037)	(0.00041)	(0.00036)
nonstop	2.22***	1.39***	2.28***
	(0.047)	(0.035)	(0.046)
lfreq_flights	0.5***		0.51***
	(0.0095)		(0.0096)
lns_origins	0.077***	0.37***	
	(0.013)	(0.015)	
prices	−0.011***	−0.0078***	−0.011***
	(0.00038)	(0.0004)	(0.00035)
rho	0.39***	0.54***	0.38***
	(0.012)	(0.0095)	(0.012)

This table presents the Nested Logit demand estimates in the 2nd stage, where we include all airline products into one nest. We adopt regular BLP instrument. The firm FEs, route FEs and quarter FEs are absorbed. The standard errors are clustered at the firm-market level.

captures the additional fuel consumed by boarding one more passenger, expressed as a share of the fuel consumption per seat of an empty flight. For the most common commercial aircraft, that share falls roughly between 15% and 30%. [5] This estimate is consistent with industry benchmarks.

First Stage Flight Frequency Cost Estimates As shown in Table 5, I estimate the parameters governing the marginal cost of operating an additional flight. The marginal cost of flight is recovered as described in subsection 5.2. The estimated coefficient on jet fuel cost per flight is positive and less than one, which is consistent with the idea that actual fuel use and paid fuel prices differ from the measured inputs. One reason is that airlines can conserve fuel

[5] Using the Boeing 737-700 as an example, its operating empty weight (OEW) is 37,648 kg and a typical seating capacity is 148. An average passenger with luggage weighs roughly 85 kg in the U.S. A common rule of thumb is that a 1% increase in weight raises fuel consumption by about 0.75% (Capehart, 2007), implying a ratio of about 25% for the Boeing 737-700.

Table 4: Supply Estimates in the Second Stage

	MC	MC	MC	MC
distance	0.034^{***}	0.035^{***}	0.034^{***}	0.041^{***}
	(0.0018)	(0.0018)	(0.0018)	(0.0017)
distance2	$2.5e-06^{***}$	$2.4e-06^{***}$	$2.3e-06^{***}$	$3.6e-06^{***}$
	(5.8e-07)	(5.8e-07)	(5.8e-07)	(5.8e-07)
nonstop	-24.34^{***}	-23.8^{***}	-21.83^{***}	-36.64^{***}
	(1.13)	(1.1)	(1.14)	(0.67)
lfreq_flights	-0.79^{***}		-1.23^{***}	-1.02^{***}
	(0.29)		(0.29)	(0.3)
lconnections	-18.62^{***}	-19.6^{***}		-17.78^{***}
	(1.74)	(1.73)		(1.74)
jfp_fuel	0.19^{***}	0.19^{***}	0.18^{***}	
	(0.016)	(0.017)	(0.017)	

This table presents the supply estimates in the 2nd stage. The firm by origin FEs and firm by destination FEs are absorbed. The standard errors are clustered at the firm-market level.

through operational adjustments such as lower cruise speeds or lighter onboard weight. Overall, the estimates suggest that the model captures the underlying cost structure of the industry reasonably well.

7 Conclusions and Plans

7.1 Conclusions

This paper studies how fuel cost shocks are transmitted into firms' operating decisions, including prices, service quality, and entry. I focus on jet fuel prices in the U.S. airline industry. I build a structural model of the airline industry featuring network effects, market power, and fuel costs, and use it to study the effects of SAF mandates — modeled as increases in effective jet fuel prices — on route networks, market power, emissions, and welfare. The model is designed to clarify how policy-induced fuel-cost shocks interact with network structure and

Table 5: Marginal Cost of Flight Frequency in the First Stage

	MC Flight
sch_time	175.8^{***}
	(50.55)
lconnections	-1677.49^{***}
	(275.45)
jfp_fuel_flight	0.87^{***}
	(0.15)
N	36328

This table presents the OLS estimates of regressing model-recovered marginal cost of flights on product features. Firm by origins/destinations and quarter FEs are absorbed. The standard errors are clustered at market by firm level.

to help guide the design of SAF policies that reduce emissions while limiting competitive distortions.

The project is of interest both academically and practically. From a research perspective, it studies cost-shock transmission in a network industry, where fuel-cost shocks can interact with network effects and market power in ways that do not arise in isolated markets. From a policy perspective, it evaluates the effects of SAF mandates in an industry where many countries and regions have already adopted or proposed similar policies, even though their downstream welfare consequences remain largely unknown.

Preliminary results suggest that jet fuel costs have significant effects on market structure and that higher fuel prices intensify market power. They also show that airline entry decisions reflect clear network considerations and that the structural estimates imply significant network spillovers in demand, marginal costs, and flight-frequency costs. Additional results will be incorporated as the estimation proceeds.

7.2 Continuing Research Plans

I will continue the estimation of entry fixed costs as described in subsection 5.2. This step is computationally demanding because entry in one market can affect profits in as many as $2 \times (C - 1) + 1$ other markets, where C is the number of cities. I plan to use the revealed-preference and moment-inequality approach in Pakes et al. (2015).

I will then conduct two sets of counterfactual analyses. The first evaluates SAF mandates, modeled as increases in effective jet fuel prices, and quantifies their effects on ticket prices, flight frequencies, route networks, market power, emissions, and overall welfare. I will also compare alternative policy designs, including differentiated SAF requirements on hub routes and alternative policy classes such as cap-and-trade schemes.

The second set of counterfactuals isolates the role of network effects by comparing the welfare effects of SAF policy in models with and without network spillovers.

References

Bardon, Paul, and Olivier Massol. 2025. "Decarbonizing Aviation with Sustainable Aviation Fuels: Myths and Realities of the Roadmaps to Net Zero by 2050." Renewable and Sustainable Energy Reviews 211 (April): 115279.

Barwick, Panle Jia, and Zhe Yuan. 2026. "Network Competition in the Airline Industry: An Empirical Framework." The RAND Journal of Economics.

Berry, Steven T. 1990. "Airport Presence as Product Differentiation." The American Economic Review 80 (2): 394–99.

Berry, Steven T., and Philip A. Haile. 2014. "Identification in Differentiated Products Markets Using Market Level Data." Econometrica 82 (5): 1749–97.

Borenstein, Severin. 1989. "Hubs and High Fares: Dominance and Market Power in the U.S. Airline Industry." The RAND Journal of Economics 20 (3): 344–65.

Ciliberto, Federico, and Elie Tamer. 2009. "Market Structure and Multiple Equilibria in Airline Markets." Econometrica 77 (6): 1791–828.

Delbecq, S., J. Fontane, N. Gourdain, T. Planès, and F. Simatos. 2023. "Sustainable Aviation in the Context of the Paris Agreement: A Review of Prospective Scenarios and Their Technological Mitigation Levers." Special Issue on Green Aviation 141 (August): 100920.

Farrell, Joseph, and Garth Saloner. 1986. “Installed Base and Compatibility: Innovation, Product Preannouncements, and Predation.” The American Economic Review 76 (5): 940–55.

He, Zhiguo, Jing Huang, and Jidong Zhou. 2023. “Open Banking: Credit Market Competition When Borrowers Own the Data.” Journal of Financial Economics 147 (2): 449–74.

Johansen, Cecilie Skov, Iva Ridjan Skov, and Alessandro Mati. 2026. “Exploring the Potential for Cost-Competitive e-SAF in European Aviation towards 2050.” Energy Reports 15 (June): 109018.

Katz, Michael L., and Carl Shapiro. 1994. “Systems Competition and Network Effects.” Journal of Economic Perspectives 8 (2): 93–115.

Klier, Thomas, and Joshua Linn. 2012. “New-Vehicle Characteristics and the Cost of the Corporate Average Fuel Economy Standard.” The RAND Journal of Economics 43 (1): 186–213.

Koopmans, Carl, and Rogier Lieshout. 2016. “Airline Cost Changes: To What Extent Are They Passed through to the Passenger?” Journal of Air Transport Management 53 (June): 1–11.

Nocke, Volker, and Nicolas Schutz. 2018. “Multiproduct-Firm Oligopoly: An Aggregative Games Approach.” Econometrica 86 (2): 523–57.

Park, Minjung. 2011. “THE ECONOMIC IMPACT OF WIRELESS NUM-

BER PORTABILITY." The Journal of Industrial Economics 59 (4): 714–45.

Sibdari, Soheil, Iman Mohammadian, and David F. Pyke. 2018. "On the Impact of Jet Fuel Cost on Airlines' Capacity Choice: Evidence from the U.S. Domestic Markets." Transportation Research Part E: Logistics and Transportation Review 111 (March): 1–17.

Wu, Mengying, Kristen McCormack, William A. Scott, Aaron Smith, Jingran Zhang, and James H. Stock. 2025. "Policy Options to Achieve US Sustainable Aviation Fuel Targets." National Bureau of Economic Research Working Paper Series No. 34326.